

Tide retrospective: identifiability from leading-part structure

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Tenth tide of the germbij arc, making its two chains structurally parallel. The 1D chain runs analyticity $\rightarrow$ growth bound $\rightarrow$ sector bound $\rightarrow$ composite $\rightarrow$ composed corollary; the multivariate chain had every link except the last. This tide composes the leading-part instantiation¹ with the multivariate composite, mirroring the 1D analytic corollary.²

2 The thing formalised

One theorem, `leading_part_pencil_difference_lower_bound` in `Laplace/Multi/LeadingIdentifiability.lean` sorry-free with zero warnings. For $L_1, L_2 \geq 0$ on ‘ $\iota \rightarrow \mathbb{R}$ ’ whose difference has a continuous homogeneous leading part P of degree m (nonzero at a direction of norm $3/2$, remainder $O(\|x\|^{m+1})$), with $L_1 + L_2 \leq C_0\|w\|^2$ on the ball of radius R and a bump ψ equal to 1 there: there exist $\kappa > 0$ and $T_0 = \max(4/R^2, u_0^{-2})$ such that for all $t \geq T_0$, under the per- t integrability premises,

$$\kappa \cdot t \cdot t^{-m-d/2} \leq \int (L_2 - L_1) \psi(e^{-tL_1} - e^{-tL_2}) dw.$$

The packaged constant is $\kappa = \text{vol}(S) \cdot c^2 e^{-4C_0}$ with S, c from the instantiation, positive by `ENNReal.toReal_pos`.

3 Proof strategy

Pure composition plus one scale calculation: the threshold T_0 makes both the window condition $4 \leq R^2 t$ and the instantiation’s scale condition $(\sqrt{t})^{-1} \leq u_0$ hold simultaneously; the latter converts through $u_0^{-1} \leq \sqrt{t}$ via `Real.sqrt_sq`, `Real.sqrt_le_sqrt`, and the inverse-comparison lemma. The final step is the multivariate composite applied verbatim, with one `ring` re-association of the packaged constant.

4 Roadblocks and resolutions

One naming iteration: `inv_le_inv_of_le` no longer exists in current Mathlib; the stable idiom for “ $a^{-1} \leq b$ iff $b^{-1} \leq a$ ” is `inv_le_comm`. Otherwise the composition compiled directly. Operationally this tide again ran concurrently with the parallel session’s tides on the repo, without interference.

¹2026-08-09-04-35-tide-germbij-leading-part.

²2026-08-09-03-25-tide-germbij-analytic-corollary.

5 What was Mathlib, what was new

Mathlib: square-root and inverse comparison lemmas, `ENNReal.toReal_pos`. New: the threshold bookkeeping and the packaging. As with the 1D corollary, the mathematical content was already merged; the value is that the multivariate chain now has the same user-facing shape as the 1D one.

6 Lessons

Renamed order-of-operations lemmas (`inv_le_inv_of_le` and `kin`) are the most common single-error class in this arc; when one fails, look for the `_comm` iff-form, which has been stable.

7 Follow-ups

- The multivariate turnkey: discharge the per- t premises from continuity and compact support, as in the 1D turnkey tide (the three integrability lemmas generalise with ‘ $\iota \rightarrow \mathbb{R}$ ’ in place of $\mathbb{R}$ almost verbatim).
- A `ContDiff`-to-remainder bridge for concrete callers.
- Note annotation: bump the pin and tag the multivariate discussion with the leading-part corollary once merged.